AI 101 for Mechanical Engineers · Module 1

Start with an engineering decision

Lesson 1.3

What you will learn

> Use the pump-inspection example to identify the decision, the measurements, the model output and the engineering action.

Lesson overview

Module 1 · Lesson 3

Start with an engineering decision

AI 101 for Mechanical Engineers

Should an inspection be requested?

Define the decision before choosing an algorithm.

1 Measurements

- Vibration (mm/s)
- Bearing temperature (°C)

Available before the decision.

→

2 Learned detector

Trained on examples; uses the measurements.

→

3 Model output

Fault score or alert.

→

4 Engineering review

Check the evidence.

→

5 Decision

Request inspection when the procedure calls for it.

A score is an estimate

Fault score = 0.8

The score has no physical unit.

It is not automatically an 80% chance of failure.

A score does not give permission to keep operating.

Define the conditions and responsibility

- **Who uses the output?**
- **Operating domain:** Where is the system intended to work?
- **Escalation path:** Who receives a concern?

Engineers check measurement quality, operating limits and model validation.

Know the limits of the example

Synthetic examples are constructed for teaching.

They do not prove that real equipment is safe.

No equipment operating limits or alert threshold are given here.

Consider both missed faults and false alarms. Keep model estimates separate from operating decisions.

AI 101 for Mechanical Engineers · Lesson 1.3

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Start with the decision, not the algorithm

- Pump vibration and bearing temperature are rising
- Question: should an inspection be requested? [11]
- Then ask how a model could help

Keep the stages separate



- Vibration in mm/s; bearing temperature in °C
- No operating limits are given in this example

Both kinds of error matter

Missed fault

- Damage can go undetected

False alarm

- Wastes inspection time
- Alerts may be acted on less readily [90]

No numerical target is set in this lesson.

A fault score of 0.8 is not 80 %

- 0.8 is a dimensionless model estimate
- It is a probability only if calibrated [12]
- Check measurement quality and model validation
- Name the user, operating domain and escalation path [9]

Different components need different checks

Component	Check
Fixed trip	Rule fires when its condition occurs
Learned detector	Performance on examples not used to fit it
Maintenance assistant	Text checked against records

Synthetic examples teach; they do not validate

- Useful for practising methods
- They do not show a real machine is safe [13]
- Say what still needs checking on real equipment

CHECK YOUR UNDERSTANDING

Name the decision, two inputs and one output in the pump example.

Answer before moving on.

CHECK YOUR UNDERSTANDING

Name the decision, two inputs and one output in the pump example.

> Decision: request an inspection? Inputs: vibration and bearing temperature. Output: a fault alert or score.

CHECK YOUR UNDERSTANDING

Does a fault score of 0.8 authorize continued operation?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a fault score of 0.8 authorize continued operation?

> No. It is an estimate. Engineers check limits, measurement quality and validation, and escalate concerns.

Key takeaways

- Define the decision before the algorithm
- Use measurements available before the decision
- Weigh missed faults and false alarms
- Keep model estimates apart from operating decisions
- Synthetic examples do not validate machinery

References

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Numbers match the course reference list. The written lesson is the authoritative text.